

Pranav Mahalingam

Ann Arbor, MI | +1 (734) 678-6093 | mpranavm@umich.edu | linkedin.com/in/pranavmahalingam | Portfolio: <https://bit.ly/mpranavm>

PROFESSIONAL SUMMARY

Software Engineer and AI Engineer with 3+ years of experience building AI/ML, computer vision, and IoT-driven systems. Proven product owner experience delivering scalable AI platforms, improving defect detection accuracy, and deploying production-grade solutions on AWS. Strong background in Python, C++, system architecture, and machine learning.

EDUCATION

University of Michigan

Master of Science in Computer Science and Engineering (MS CSE)

Ann Arbor, MI

Aug. 2025 – Present

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science

Chennai, India

July 2018 – May 2022

TECHNICAL SKILLS

Languages: Python, C++

IoT/Embedded: ESP32, Raspberry Pi, Arduino

AI/ML: Deep Learning, Computer Vision, NLP

Cloud/Tools: AWS, PySpark, Streamlit, Git, Docker

PROFESSIONAL EXPERIENCE

Student Life Technology Solutions, University of Michigan

Computer Consultant 1 (Part-time)

Ann Arbor, MI

Jan. 2026 – Present

- Supported and maintained distributed infrastructure systems across 75+ campus buildings, including AV, digital signage, and POS systems.
- Managed inventory lifecycle for 10,000+ devices and led coordinated infrastructure upgrades to ensure system availability.

Dhvani Analytic Intelligence

AI Engineer

Chennai, India

Sept. 2024 – Aug. 2025

- Served as Product Owner for the PixIQ AI system, leading a cross-functional applications team to deliver 5+ customized AI solutions for diverse industries and reducing project turnaround time by 15%.
- Designed and implemented scalable AI system architectures (Python, AWS) for industrial floor deployment, increasing operational throughput by 25%.
- Optimized solution reliability through rigorous prototype validation and detailed code reviews to align with client specifications, increasing defect detection precision by 12%.

Dhvani Analytic Intelligence

Junior AI Engineer

Chennai, India

Feb. 2023 – Aug. 2024

- Delivered production-grade computer vision solutions for enterprise clients including GE Healthcare, improving defect detection accuracy and operational efficiency.

Vaayusastra Aerospace / Freelance

AI & IoT Trainer

Chennai, India

Mar. 2019 – July 2025

- Mentored 700+ students through hands-on AI/IoT workshops and intensive 12-day internship programs covering hardware and software.

RESEARCH EXPERIENCE

National Institute of Technology

Research Intern

Trichy, India

Oct. 2021 – Apr. 2022

- Developed an NLP-based automated MCQ generation system using LSTM and attention mechanisms, achieving 96.7% recall.
- Published results at the I3CS 2021 conference.

KEY PROJECTS

BudgetBruh | TartanHacks 2026 (CMU)

Behavioral Finance App

Pittsburgh, PA

Feb. 2026

- Developed an LLM-powered behavioral finance app that recontextualizes spending into labor hours and carbon impact, featuring a "Freedom Cost" algorithm to visualize how discretionary purchases delay user-defined milestones.

Git-Landscaper | Open Source Innovation

University of Michigan

Ann Arbor, MI

Jan. 2026

- Developed a GitHub extension to automate repository organization by identifying and removing dead branches, significantly improving version control hygiene for users.

Git-Hired | AI Virtual Interviewer

Best Project – Claude Builders Club Hackathon

Ann Arbor, MI

Nov. 2025

- Built an AI-powered virtual interviewer using agentic AI concepts to simulate real interview scenarios within a 24-hour hackathon.

PUBLICATIONS

Automated Multiple Choice Question Creation Using Synonymization and Factual Confirmation, I3CS 2021.